MAKING BENELUX CITY POPULATION DATA FAIR

FAIR Data Engineering

Presented by Group - 15

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THE DATASET

What: UN Statistics Division city population time series, Benelux subset: 531 rows, 46 cities, 2000 to 2023

Found: DataHub (datahub.io/core/population-city), republishing UNdata. Licence: PDDL 1.0 (public domain)

Original format: Single CSV, 11 columns. Unnecessary free text

Existing metadata: only on the DataHub web page and its datapackage.json (title, description, licence, source). Nothing in the CSV, no persistent identifier, no contact

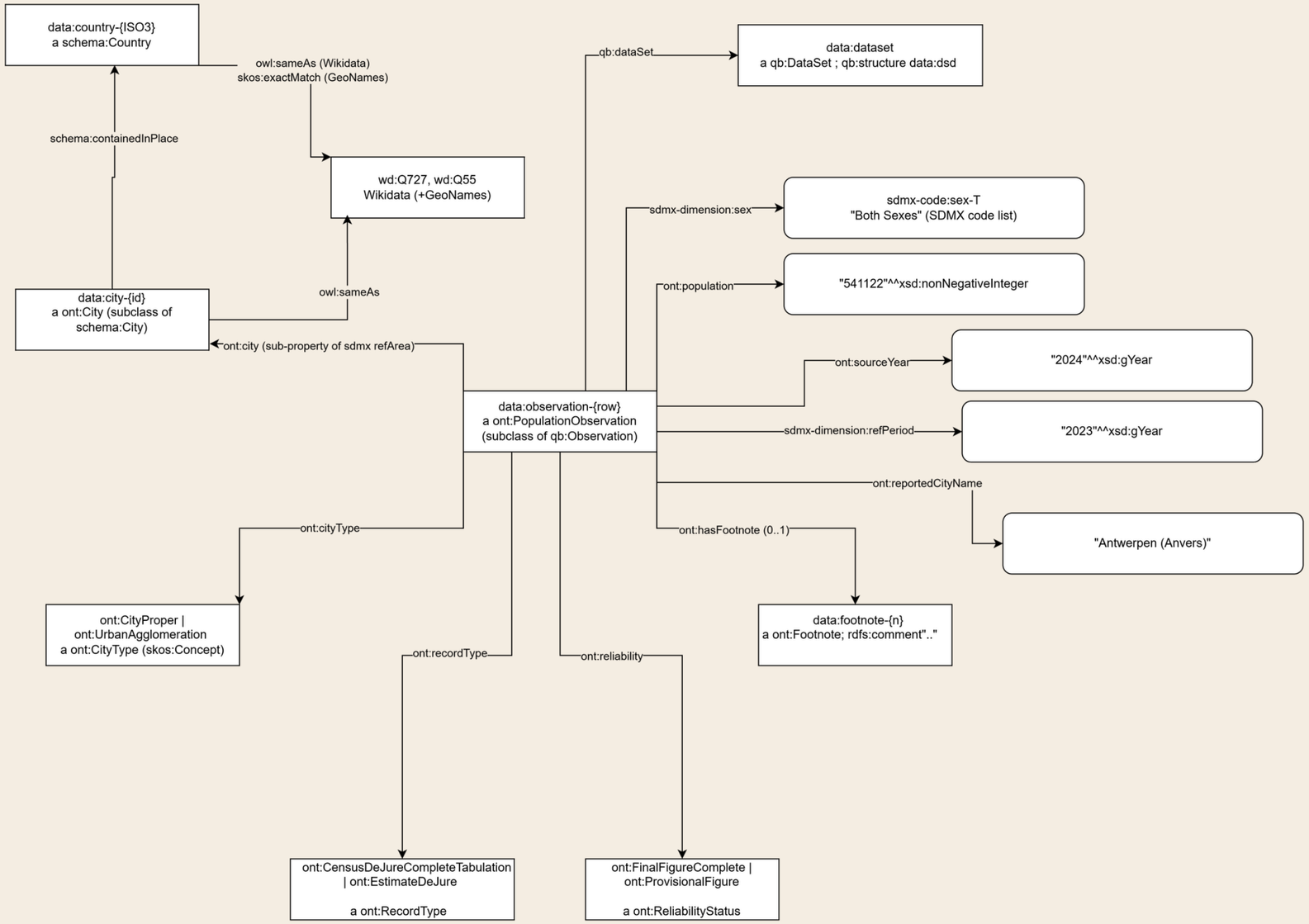
FAIRIFICATION PLANNING

- **Goals**
  - W3ID identifiers and rich metadata in a FAIR Data Point (F, A)
  - RDF with shared vocabularies and Wikidata links (I)
  - Licence and provenance (R)
- **Stakeholders**
  - Us: Data preparation; metadata; modelling; RDF creation and data publication
  - Luiz Bonino da Silva Santos: FAIR expert
  - United Nations Statistics Division: Data provider
- **Reuse scenarios**
  - Urban planner: How did Dutch cities grow between 2003 and 2015?
  - Demographer: How much larger is the agglomeration than the city proper?
  - Journalist or student: What is each city's population density?
  - Wikidata editor: Do the UN figures match Wikidata's?

DATA PREPARATION

- Separated the Benelux data
- Moved footnotes from the bottom of the data to an actual column, then stored each one and linked to them
- Ignored columns that were always the same value
- Some city names needed to have their clarity improved
- Reconciled cities and countries with WikiData

SEMANTIC DATA MODEL

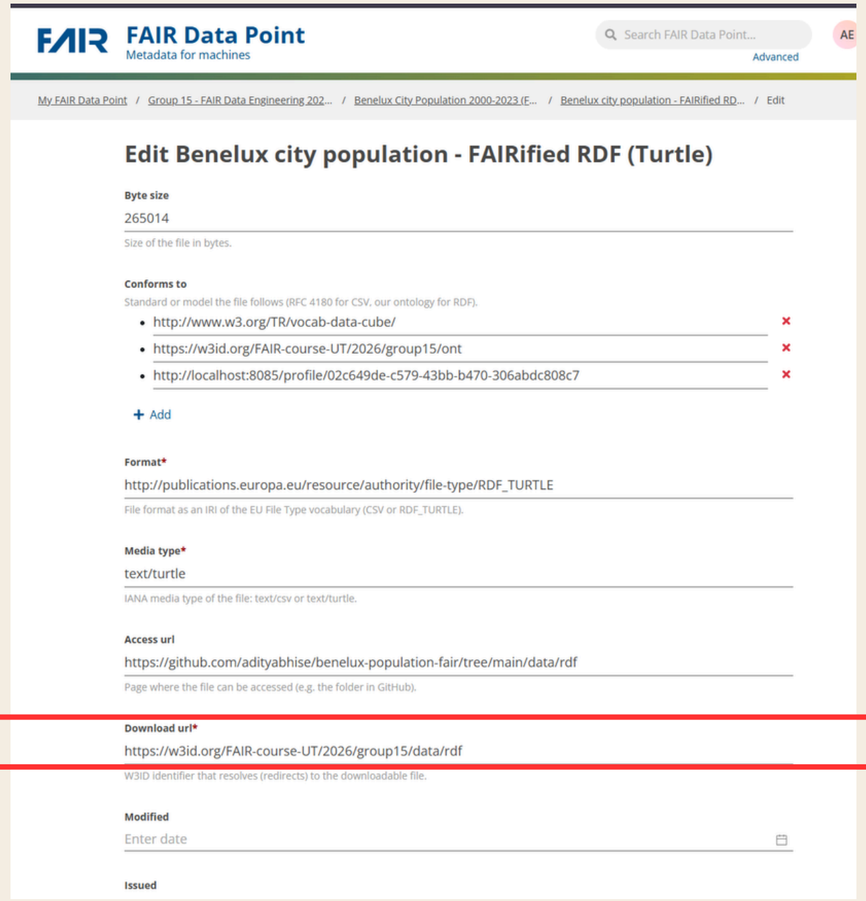


- RDF Data Cube: city, city type, method, year and sex are dimensions; population is the measure
- Reused vocabularies: Data Cube, SDMX, schema.org, SKOS. Own terms only where missing
- Persistent IRIs (F1)
- Linked (I3): cities and countries owl:sameAs Wikidata, countries also to GeoNames

METADATA SCHEMA

Property	Rule	FAIR
dct:identifier	exactly 1, must start with https://w3id.org/	F1
dcat:keyword	at least 3	F2, F4
dct:spatial, dct:temporal	required, as IRIs (Wikidata countries, year interval)	F2, I2
prov:wasDerivedFrom, dct:source	required: DataHub file and UNSD	R1.2
dct:conformsTo	required: our ontology and RDF Data Cube	R1.3
dcat:contactPoint, dct:creator	required	R1
dcat:downloadURL (distribution)	must be a w3id address	F1, A1

METADATA SCHEMA IN ACTION



Form with our w3id download URL

Unable to update entity data.

View report

```
[] a sh:ValidationReport;
  sh:conforms false;
  rdf4j:truncated false;
  sh:result [ a sh:ValidationResult;
      sh:focusNode <http://localhost:8085/distribution/809d4354-3039-4025-a240-462fcb3df2b6>;
      rsx:shapesGraph rdf4j:SHACLShapeGraph;
      sh:value <https://github.com/adityabhise/benelux-population-fair/raw/main/data/rdf/benelux-cit
      sh:resultPath dcat:downloadURL;
      sh:sourceConstraintComponent sh:PatternConstraintComponent;
      sh:resultSeverity sh:Violation;
      sh:resultMessage "The download URL must be the group's W3ID identifier"@en;
      sh:sourceShape [ a sh:PropertyShape;
          sh:message "The download URL must be the group's W3ID identifier"@en;
          sh:name "Download URL";
          sh:description "W3ID identifier that resolves (redirects) to the downloadable file.";
          sh:path dcat:downloadURL;
          sh:pattern "^https://w3id.org/FAIR-course-UT/"
        ]
    ] .
```

GitHub link entered: the FDP refuses to save

- The rule: dcat:downloadURL must match https://w3id.org/FAIR-course-UT/
- The FDP validates every record with SHACL and returns a report: field, wrong value, rule broken and our own message
- Only persistent identifiers can enter the metadata (F1, A1)

FAIRNESS IMPROVEMENTS

Before	After
No identifier and lackluster metadata	Persistent W3ID identifier and DCAT metadata in the FAIR Data Point
A download link that might change	Identifiers resolve over HTTPS through W3ID and the metadata is available as RDF
Free text names and cells with mixed languages	RDF with Data Cube, SKOS and schema.org, Linked to WIkidata and GeoNames
The data license and source were only finable on a webpage	A machine-readable license, data provenance and our own ontology

THANK YOU
QUESTIONS ?